

FITTED PARAMETERS AND RESULTS

Reservoir Water-Level Cubic Model

1. Fitted Parameters

Parameter	Fitted Value	Standard Error	t	p
a (m/h ³)	-0.00010595068	8.902×10^{-6}	-11.9	9.08×10^{-27}
b (m/h ²)	0.0096061241	0.0009718	9.885	5.393×10^{-20}
c (m/h)	-0.091243028	0.02998	-3.044	0.002555
d (m)	14.150575	0.2479	57.08	1.194×10^{-157}

2. Fitted Equation

$$h(t) = -0.00010595068t^3 + 0.0096061241t^2 - 0.091243028t + 14.150575$$

3. Model Statistics

Statistic	Value	Unit / Meaning
SSE	322.4758	m ²
R ²	0.8414634	Coefficient of determination
s	1.065588	m; standard error of estimate
Degrees of freedom	284	288 observations - 4 parameters

4. Area

Quantity	Result	Unit
Area under fitted curve	1261.19984	m·h
Quadrature error	$\pm 1.40 \times 10^{-11}$	m·h

5. Reading of the Residuals

The residuals are centered around zero, with a mean of $+1.31 \times 10^{-14}$ m. There is no material gradual increase or decrease in the residuals. The residual spread does not clearly increase as the water level rises. The largest absolute residual is 2.7825 m (278.25 cm), which is substantially above the 1 cm logger accuracy. Overall, the residuals indicate that the cubic model provides a reasonably good fit, as supported by its R² of 0.8415, although some relatively large deviations remain.